

Model Question Paper-01

COURSE- B.Tech

DEPTT. – Applied Sciences

YEAR: - 1st (Second SEM)

SUBJECT NAME: -Engineering Physics (BAS201)

TIME: 3.00 Hours

TOTAL MARKS: 70

Note: Answer all questions. Draw neat sketch, wherever necessary. Suitably assume missing data, if any.**Section A****Q1. Attempt all the parts.****[2X7=14]**

a.	What is the physical significance of wave function?	2	[CO1], K1
b.	Calculate the skin depth for a frequency of 10^{10} Hz for silver. The conductivity for silver is 2×10^7 S/m and $\mu = 4\pi \times 10^{-7}$ H/m	2	[CO2], K3
c.	What is Rayleigh's criterion of resolution?	2	[CO3], K2
d.	What do you understand by resolving power of a grating?	2	[CO3], K1
e.	What is spontaneous and stimulated emission of radiation?	2	[CO4], K1
f.	What do you mean by metastable state?	2	[CO4], K1
g.	What is quantum dot, quantum wire and quantum well?	2	[CO5], K1

Section B**Q2. Attempt any three of the following:****[7X3=21]**

a.	What do you mean by Compton effect? Derive relation for Compton shift.	7	[CO1], K2
b.	Explain the concept of displacement current and show how it led to the modification of Ampere's law.	7	[CO2], K3
c.	Describe Fraunhofer single slit diffraction and deduce the position of maxima and minima. Show that the relative intensities of maxima and minima are nearly $1; \frac{1}{22}; \frac{1}{61}; \frac{1}{121}; 1; \frac{1}{22}; \frac{1}{61}; \frac{1}{121}$.	7	[CO3], K2
d.	Explain acceptance angle, acceptance cone of an optical fiber. What do you mean by numerical aperture? Derive expressions for them.	7	[CO4], K2
e.	Discuss type – I and type – II superconductors. A superconducting tin has a critical temperature 3.7 K at zero magnetic fields and a critical field of 0.0306 Tesla at 0K. Find the critical field at 2K.	7	[CO5], K4

Section C**Q3. Attempt any one part of the following:****[7X1=7]**

a.	Derive time independent and time dependent Schrodinger wave equation.	7	[CO1], K2
b.	What is phase and group velocity? Derive expression for phase and group velocity and establish relation between them.	7	[CO1], K2

Q4. Attempt any one part of the following:**[7X1=7]**

a.	The sunlight strikes the upper atmosphere of the Earth with energy flux 1.38 KW/m^2 . What will be the peak value of electric and magnetic fields at that point	7	[CO2], K3
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b.	Write down the Maxwell equation in differential form for free space. Prove that the electromagnetic waves are transverse in nature.	7	[CO2], K2
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Q5. Attempt any one part of the following:

[7X1=7]

a.	Describe and explain the formation of Newton's rings in reflected monochromatic light. Newton's rings are observed normally in reflected light of wavelength 6000 Å. The diameter of 10 th dark ring is 0.50 cm. Find the radius of curvature of the lens and thickness of the film.	7	[CO3], K3
b.	Discuss the phenomena of interference of light in thin films and find the condition of maxima and minima for reflected light.	7	[CO3], K2

Q6. Attempt any one part of the following:

[7X1=7]

a.	Discuss different types of optical fiber. Calculate a fractional difference between core and cladding refractive indices for a step index fiber having core and cladding refractive indices 1.55 and 1.50 respectively. Also calculate acceptance angle, numerical aperture and the critical angle.	7	[CO4], K3
b.	Explain the construction and working of Ruby Laser with a neat and clean diagram.	7	[CO4], K2

Q7. Attempt any one part of the following:

[7X1=7]

a.	What is Meissner's effect? Show that material in superconducting state is perfect diamagnetic in nature.	7	[CO5], K2
b.	What are nano materials? Describe the chemical vapour deposition technique for synthesis of nano materials.	7	[CO5], K2